Stack

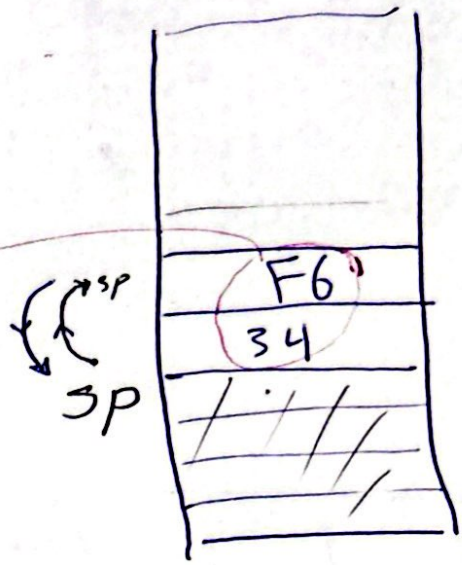
لا بد
Register + 16

* Push AX

copy

AX دخول
stack ل

ex AX = 34F6



* Pop CX

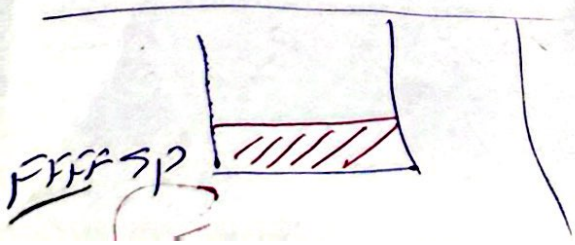
قدر

CX = 34F6

* after any Push ⇒ SP dec. by 2 → +2

" " Pop ⇒ SP Inc by 2 → +2

* offset add. on stack of 1st location
nerdy to Push data



→ SP − 1

≡ FFFE

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(6)

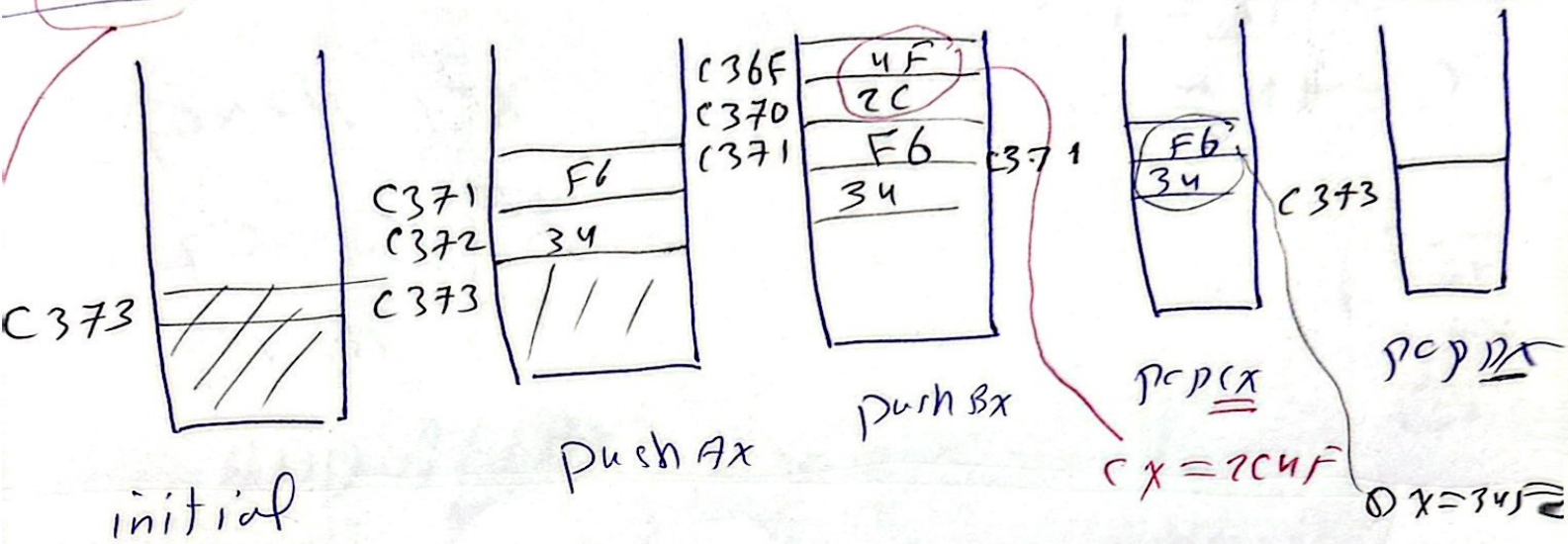
AX = 34F6

BX = 2C4F

SP = C373

* Push AX N.C
* Push BX N.C
* pop CX
* pop DX

show stack



what value

LIFO

AX = 34F6

BX = 2C4F

CX = ~~34F6~~ 2C4F

DX = ~~2C4F~~ 34F6

SP = C373 ~~- 2 - 2 + 2 + 2~~ = C373

(7th) MOV AX, 34F6

MOV BX, C342

push AX
pop BX

Value

AX = 34F6

BX = 34F6

(7th)

SP = 123A

AX = 462A

DI = 872C

DX = 4F5D

push DX

pop BX

push AX

ADD AX, BX

+

push AX

pop DX

AX

AX

DX

BX = 4F5D

462A
4F5D +
9577

AX = 9577

DX = 9577

AX = 9577

BX = 4F5D

DX = 9577

SP = 123A $\boxed{-2}$ +2 ~~-2~~ ~~-2~~ ~~+2~~ = $\boxed{1238}$

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MOV AL, 8C

MOV BL, 7E

ADD AL, BL

AL = 8C

BL = 7E +

1 0 A

AL = 0A

AL = 7E

CF = 1

(A)16 → BH

MOV BH, 0AH

MOV AX, -4

AX = FFFC

+4

0000 0000 0000 0100

1111 1111 1111 1011 +1

1111 1111 1111 1100

VIP

illegal اسباب

① size

*Mov AL, 4F ✓

*Mov AX, 4F3C ✓

Mov AL, 4F2 X

Mov AX, 00uC ✓

الغلط size

* Mov AL, BL ✓

Mov AX, BX ✓

Mov AL, BX X

Mov AX, BL X

Register Mode $\xrightarrow{\text{must}}$ equal size

---

* Mov AL, [BX] ✓

Mov AX, [BX] ✓

Mov AL, [BL] X

offset ⑯

Mov [AX], [BX] X

no mov bet^n 2 memory location

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(X) MOV DS, $\boxed{34F6}$ 16 X

DS	↔ ✓	AX
CS		BX
CS		CX
SS		DX

segment Registers

accept mov from general Registers (16)
(AX, BX, CX, DX)

Push
pop
AL
CS
AH
α

Register (16)

MOV AX, [BP + BX + 30] α
[SI + DI + 30] α

no Base - Base Mode
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